De La Salle University

College of Science

Department of Physics

COURSE CODE: SEMPY01

COURSE TITLE: SEMICONDUCTOR PHYSICS

Instructor: RAGASA, EUGENE JOSEPH. M.
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Consultation Hours: `
Class Schedule: TF 1245-1415 -

Room/ Recurring Zoom link:

TYPE OF COURSE		
_X_ Hybrid	__ Fully online	__ Fully Asynchronous
__ In- Person	__ Predominantly Online	__ Predominantly In- person

GENERAL COURSE DESCRIPTION

The course deals with the physics of semiconductors and semiconductor devices. It includes behavior of electrons moving in a periodic potential and the features of the energy band structure that distinguish semiconductors from metals and insulators; the statistics of electrons and holes in semiconductors in equilibrium, behavior of excess carriers in semiconductors; properties of a pn junction, the transistor, LEDs, semiconductor lasers, and other semiconductor devices.

SPECIFIC COURSE DESCRIPTION

This course develops students' understanding of semiconductor physics and devices. Students will learn to analyze carrier behavior, interpret band structures, and apply statistical models to semiconductor materials. By the end of the course, they will be able to explain and evaluate the operation of devices such as pn junctions, transistors, LEDs, and lasers.

TARGET SUSTAINABLE DEVELOPMENT GOALS (SDG)

This course aligns with United Nations Sustainable Development Goal 4: Ensure inclusive equitable quality education and promote lifelong learning opportunities for all. Specifically, Goal 4.4: By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship. Upon completion of this course, students will be equipped with the skills necessary for the academe and industry.

EXPECTED LASALLIAN GRADUATE ATTRIBUTES (ELGA)

ELGA A	Culturally Sensitive Critical and Creative Thinker
ELGA B	Effective Communicator
ELGA C	Reflective Lifelong Learner and Competent Self- Nurturer**
ELGA D	Engaged and Service-Driven Citizen
ELGA E	Virtue Guided Decision-Maker
ELGA F	Technically Competent Professional

PROGRAM LEARNING OUTCOMES (PLO)

PLO 1	apply mathematical, computational and experimental methods in solving physical problems*
PLO 2	able to evaluate quality of information gathered from varied sources
PLO 3	capable of translating scientific knowledge and methods into innovations in materials science*
PLO 4	synthesize and effectively communicate scientific information*
PLO 5	able to express thoughts in a logical, clear, concise, and precise manner*
PLO 6	apply scientific reasoning to arrive at decisions*
PLO 7	actively pursue new knowledge, be open to new ideas, and respect other people's viewpoints*
PLO 8	employ scientific skills and knowledge for the improvement of human life and the preservation of the environment*
PLO 9	uphold intellectual honesty and integrity in their conduct
PLO 10	pursue personal and professional growth by constantly acquiring new skills and keeping abreast with technological advancement*

*CHED competency standards for BS Physics/BS Applied Physics (CHED Memo No. 13, Series of 2024, Article IV, Section 6)
** United Nations Sustainable Development Goal (UNSDG under Goal 4: Quality Education)

NEW PROGRAM LEARNING OUTCOMES (PLO)	
PLO 1	Apply fundamental principles of physics, mathematics, computation, and interdisciplinary knowledge to analyze and solve complex problems.*
PLO 2	Critically evaluate scientific, technical, and economic information from varied sources to design and develop innovative and relevant applications.
PLO 3	Synthesize and communicate complex scientific information and arguments effectively in both oral and written forms to diverse audiences in a logical, clear, and precise manner.*
PLO 4	Uphold intellectual honesty and professional integrity while employing scientific knowledge and skills for the improvement of human life and the preservation of the environment, in appreciation of God's Creation.*
PLO 5	Demonstrate a commitment to lifelong learning by independently acquiring new knowledge, keeping abreast of technological advancements, and engaging respectfully with new ideas and diverse viewpoints.*

COURSE LEARNING OUTCOMES (CLO)	
CLO1	Analyze and predict the behavior of charge carriers and explain the operation of basic semiconductor devices using energy band theory and transport models.
CLO2	Apply logical and systematic problem-solving strategies to quantitative and conceptual problems.
CLO3	Communicate scientific reasoning and solutions clearly in both written and oral formats.
CLO4	Relate semiconductor physics concepts to real-world electronic and optoelectronic applications.

CLOs	SDG Goals																
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
CLO1	IC	IC	IC	DC	IC	IC	IC	IC	DC	IC	IC	IC	IC	IC	IC	IC	IC
CLO2	IC	IC	IC	DC	IC	IC	IC	IC	DC	IC	IC	IC	IC	IC	IC	IC	IC
CLO3	IC	IC	PC	DC	IC	IC	IC	IC	IC	IC	IC	IC	PC	IC	IC	DC	IC
CLO4	IC	IC	IC	PC	IC	IC	IC	IC	DC	IC	DC	IC	PC	IC	IC	IC	PC

Code for the Course Contribution to the SDGs		Code Description
IC	Indirectly contributes	The course has minimal or indirect contribution to the achievement of the SDG because another SDG is more directly embedded in it or it is not intended to directly address any of the SDGs.
DC	Directly Contributes	The course directly contributes to the achievement of the SDG through its projects/activities.
PC	Potentially Contributes	The course is not intended to address any of the SDGs but its projects/activities have the potential to contribute to the SDG.

CLOs	ELGA A	ELGA B	ELGA C	ELGA D	ELGA E	ELGA F
CLO1	I, E	-	-	-	-	I, E
CLO2	E, D	E	E	-	E	D
CLO3	E	-	E	-	D	E
CLO4	E, D	E	E	E	E	E

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10
CLO1	I, E	-	-	-	-	E	-	-	-	E
CLO2	E, D	E	-	E	E	D	-	-	E	E
CLO3	-	D	-	-	-	-	E	E	D	E

CLO4	E	E	-	E	E	E	E	E	E	E

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5
CLO1	I, E	–	–	–	E
CLO2	E, D	E	E	E	E
CLO3	–	–	–	D	E
CLO4	E	E	E	E	E

Code for developmental levels (CLO-ELGA and CLO-PLO matrices)		Code Description Note: a PLO may have multiple codes if it addresses the outcome at different levels.
I	Introductory	Students are introduced to a new concept or skill
E	Enabling	Enhancing, enriching, and improving what is introduced
D	Demonstrative	Demonstrating the knowledge of skill for assessment

FINAL COURSE OUTPUT

As evidence of attaining the above learning outcomes, the student must do and submit the following during the indicated dates within the term.

Learning Outcome	Required Output	Due Date
CLO1-4	Portfolio of assignments/problem sets/seat works	Week 3, 7, 10
CLO1-3	Long Exam 1 and Long Exam 2	Week 5, Week 10
CLO1–4	Final Exam	Week 14

RUBRIC FOR ASSESSMENT

The following rubrics will be used for the portfolio of assignments/problem sets/seat works.

CRITERIA	EXEMPLARY	SATISFACTORY	DEVELOPING	BEGINNING
Grade	**4**	**3**	**2**	**1**
Organization (20%)	Problem set is well-organized and structured.	Problem set is organized but lacks certain key elements.	Problem set show organization but has several portions that are not relevant.	No adequate understanding of the topic
Scientific Accuracy (50%)	Scientific explanations or facts presented/cited are 100% accurate.	Scientific explanations or facts presented/cited show some inaccuracies.	Scientific explanations or facts presented/cited show a significant number of inaccuracies.	Scientific explanations or facts presented or cited are all misconceptions.
Presentation of Arguments or Explanations (30%)	Arguments and explanations presented are clear, valid, and convincing.	Arguments presented are clear, valid, and convincing but has several flaws.	The arguments and explanations presented only partially addressed the problem.	The arguments and explanations presented do not in any way address the problem.

OTHER REQUIREMENTS AND ASSESSMENTS

Aside from the assignments, problem sets, and seat works the students will be assessed at other times during the term by the following: Two Long exams and a Final Exam.

GRADING SYSTEM

*The average of the long exams will be compared with the final exam. Whichever is higher gets 50% of the final grade; the other gets 35%.

Two (2) Long Exams	50% or 35%*
Final Exam	50% or 35%*
Portfolio of Assignments/Problem Sets/ Seat Works & Teacher's Evaluation	15%
TOTAL:	100%
Passing Grade:	50%

Scale:

94-100%	**-**	**4.0**
87-93%	**-**	**3.5**
80-86%	**-**	**3.0**
73-79%	**-**	**2.5**
66-72%	**-**	**2.0**
58-65%	**-**	**1.5**
50-57%	**-**	**1.0**
<50%	**-**	**0.0**

GENERATIVE AI-USE POLICY

AI Usage Policy Table

Grade Component	Usage Policy Level	Notes
Assignments, problem sets, seat works	(3) Banned	No AI assistance is permitted during any form of assessment that evaluates individual understanding.
Exams / Quizzes	(3) Banned	No AI assistance is permitted during any form of assessment that evaluates individual understanding.
Technical Report & Presentation	(1) Free to use	AI tools (e.g., for slide formatting, grammar editing, or chart layout) may be used with attribution.

AI Usage Policy Statement

This course offers a focused study on the physical principles governing semiconductors and their applications in electronic and optoelectronic devices. Core learning objectives include analyzing carrier behavior using band theory, applying transport and statistical models, and explaining the operation of fundamental semiconductor devices.

While generative AI can support learning by providing visualizations, simplifying complex concepts, and offering guided problem-solving strategies, excessive reliance on AI-generated solutions may hinder the development of analytical thinking and a deep understanding of device physics.

Therefore, in this course, students are allowed to use generative AI tools only to support idea generation, or reinforce concepts through self-study. These tools may be used as references for exploratory learning or for generating visual aids in presentations, provided proper attribution is given.

Students are not allowed to use generative AI to complete graded problem sets, derivations, or written explanations that are meant to assess individual understanding. Direct copying of AI-generated solutions or texts in these contexts constitutes academic dishonesty.

All allowed usage must be transparent, cited, and aligned with academic integrity policies. Violations will be treated as academic dishonesty.

LEARNING PLAN

LEARNING OUTCOME	TOPIC	WEEK NO.	REQUIRED READINGS and/or ONLINE MATERIALS/LINKS	LEARNING ACTIVITIES
LEARNING OUTCOME (CLO)	**TOPIC**	**WEEK NO.**	**REQUIRED READINGS and/or ONLINE MATERIALS/LINKS**	**LEARNING ACTIVITIES**
CLO1- 4	Introduction to the Quantum Theory of Solids Formation of energy bands, Kronig-Penney model, Electron effective mass, Concept of holes Metals, insulators, semiconductors, Density of states function, Fermi-Dirac probability function, Distribution function and the Fermi Energy	Weeks 1-2	Neamen, D. An Introduction to Semiconductor Devices, (Latest Edition) Chapter 3	Lecture and Discussions 4th Hour Activity: Problem Solving
CLO1-4	Semiconductors in Equilibrium Equilibrium distribution of electron and holes, Intrinsic semiconductors, Extrinsic semiconductors, Statistics of donors and acceptors	Weeks 3-4	Neamen, D. An Introduction to Semiconductor Devices, (Latest Edition) Chapter 4	Lecture and Discussions 4th Hour Activity: Problem Solving
CLO1-3	Exam 1	Week 5		In-person exam
CLO1-4	Carrier Transport Phenomena Carrier drift, Carrier diffusion, Mobility, Diffusion coefficient, Graded impurity distribution Nonequilibrium Excess Carriers in Semiconductors Carrier generation and recombination, Mathematical analysis of excess carriers, Ambipolar transport, Quasi Fermi levels, Excess carrier lifetimes	Weeks 6–7 Weeks 8–9	Neamen, D. An Introduction to Semiconductor Devices, (Latest Edition) Chapter 5 Neamen, D. An Introduction to Semiconductor Devices, (Latest Edition) Chapter 6	Lecture and Discussions 4th Hour Activity: Problem Solving

CLO1-3	Exam 2	Week 10		In-person exam
CLO1-4	The pn Junction Basic structure, Zero applied bias, Reverse applied bias, Forward bias, I-V characteristics Metal-Semiconductor Contacts Potential barrier Schottky barriers Ohmic contacts Semiconductor Devices Optoelectronic devices, Semiconductor power devices	Week 11-13	Neamen, D. An Introduction to Semiconductor Devices, (Latest Edition) Chapter 7 Neamen, D. An Introduction to Semiconductor Devices, (Latest Edition) Chapter 9 Neamen, D. An Introduction to Semiconductor Devices, (Latest Edition) Chapters 14 & 15	Lecture and Discussions 4th Hour Activity: Problem Solving
CLO1-3	Final Exam	Week 14		In-person exam

TEACHING STRATEGIES

The course will use active learning through problem-solving, cooperative learning via group work and peer discussions, and inductive approaches such as problem-based learning, round table discussions, and whole-brain teaching to support engagement and deeper understanding.

REFERENCES

- *Neamen, D. Semiconductor Physics and Devices: Basic Principles, Latest Edition, McGraw-Hill.*
- *Neamen, D. An Introduction to Semiconductor Devices, Latest Edition, McGraw-Hill.Rudden, M. N. (Latest Edition). Elements of Solid State Physics, John Wiley & Sons.*
- *Pierret, R. Advanced Semiconductor Fundamentals vol.VI, Modular Series on Solid State Devices. Purdue University.*
- *Pierret, R. Semiconductor Device Fundamentals. Addison-Wesley Publishing Company*
- *Van Zeghbroeck, B. Principles of Semiconductor Devices, Latest edition. University of Colorado. Website http://ece-www.colorado.edu*
- *Kittel. Introduction to Solid State Physics, 8th Ed, 2004, John Wiley & Sons.*
- *Sze. Physics of Semiconductor Devices, Latest edition, Wiley International.*
- *Abramowitz and Stegun. Handbook of Mathematical Functions, Latest edition, Dover*
- *Arfken and Weber. Mathematical Methods for Physicists, Latest edition Academic Press*
- *Spiegel. Mathematical Handbook of Formulas and Tables, Latest edition. McGraw-Hill*
- *AK Lectures. 2021. Solid State Physics. https://aklectures.com/subject/modernphysics/chemical-bonds-and-solid-state-physics*
- *University of Cambridge, 2020. Introduction to Semiconductors. https://www.doitpoms.ac.uk/tlplib/semiconductors/index.php*
- *Markvart and Castaner. 2012. Chapter IA-2: Semiconductor Materials and Modelling. Practical Handbook on Photovoltaics, Second ed. Elsevier. Accessible: https://www-sciencedirectcom. dlsu.idm.oclc.org/science/article/pii/B9780123859341000027#s0015*

CLASS POLICIES

Attendance and Participation

Regular attendance is expected. Active participation in discussions is part of your course engagement score. Notify the instructor in advance if you will miss a scheduled class or submission.

Submission of Requirements

All assignments must be submitted via the designated learning platform (e.g., AnimoSpace or Google Classroom) before the deadline.

Late submissions will incur a 5% penalty per day, up to a maximum of 3 days. Submissions beyond that will not be accepted without valid justification. Requests for deadline extensions must be submitted in writing at least 24 hours in advance and are subject to instructor approval.

Academic Integrity and Plagiarism
Plagiarism, fabrication of data, and unauthorized collaboration will not be tolerated.
Violations will be reported to the university's academic integrity office and may result in a failing grade for the assignment or course.

Use of Generative AI Tools
Refer to the official AI Usage Policy included in the syllabus. AI tools may be used for specific components such as code debugging or visual design but not for generating entire reports or exam responses. Misuse of AI tools (e.g., passing AI-generated output as original work) is considered academic dishonesty.

Group Work and Collaboration
Group projects must reflect equitable contributions. Peer evaluation will be used to assess individual participation. You are encouraged to discuss course concepts with classmates but must submit your own original work unless explicitly instructed otherwise.

Consultation and Communication
Students are encouraged to reach out during consultation hours or via email/official platform messaging for academic concerns. Response time for emails is typically within 48 hours during working days.

Professionalism and Respect
Treat your peers, instructors, and guest speakers with professional courtesy and respect.
Discriminatory, disruptive, or disrespectful behavior will not be tolerated in class discussions, group work, or online communication.

Emergency or Special Circumstances
If you encounter personal, medical, or family-related issues that affect your academic performance, please contact the instructor as soon as possible.
Arrangements can be made under compassionate grounds, aligned with university policies.

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